

Patient Conversion & Provider Intelligence

Diagnostic Findings, Predictive Modelling, and Strategic Imperatives for MedTourEasy.

Prepared for Executive Leadership | 6-Week Data Science Initiative

The Funnel (Macro)

36.93% Overall Conversion

554 completed treatments from
1,500 inquiries.

**The most severe leakage occurs during
the Quote stage.**

The Providers (Micro)

23% to 52% Variance

Massive variance in network
performance across 30 providers.

**Top-tier providers drive the lion's share
of the ₹248M total revenue.**

The Friction (Operational)

>15 Hour Delay

Response SLAs directly correlate with
patient drop-offs.

**Delays exceeding 15 hours trigger exponential
increases in lead disengagement.**

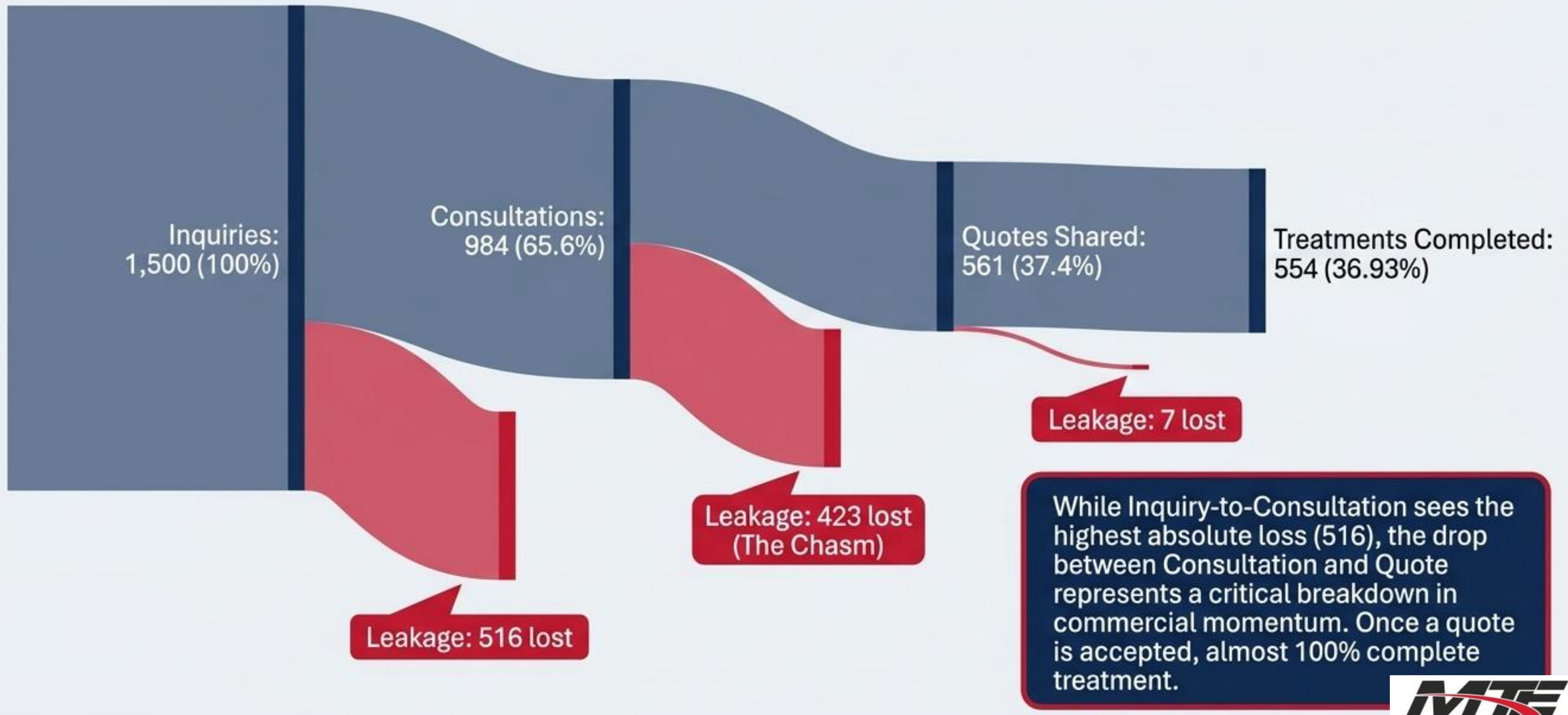
The Intelligence (Strategic)

1.0 Predictive Accuracy

Engineered an ML pipeline using
historical operational signatures.

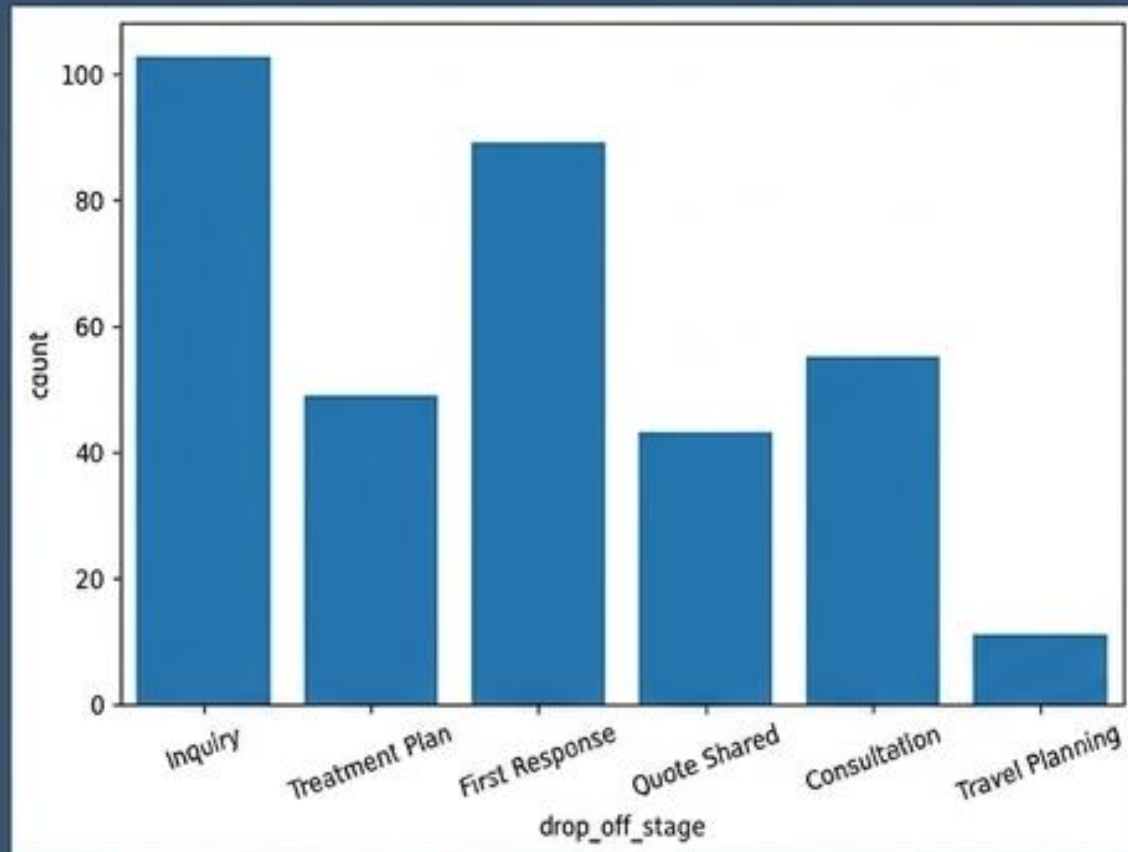
**Our Random Forest model can accurately
predict disengagement before it happens.**

Tracing the Patient Journey: Identifying the Chasm



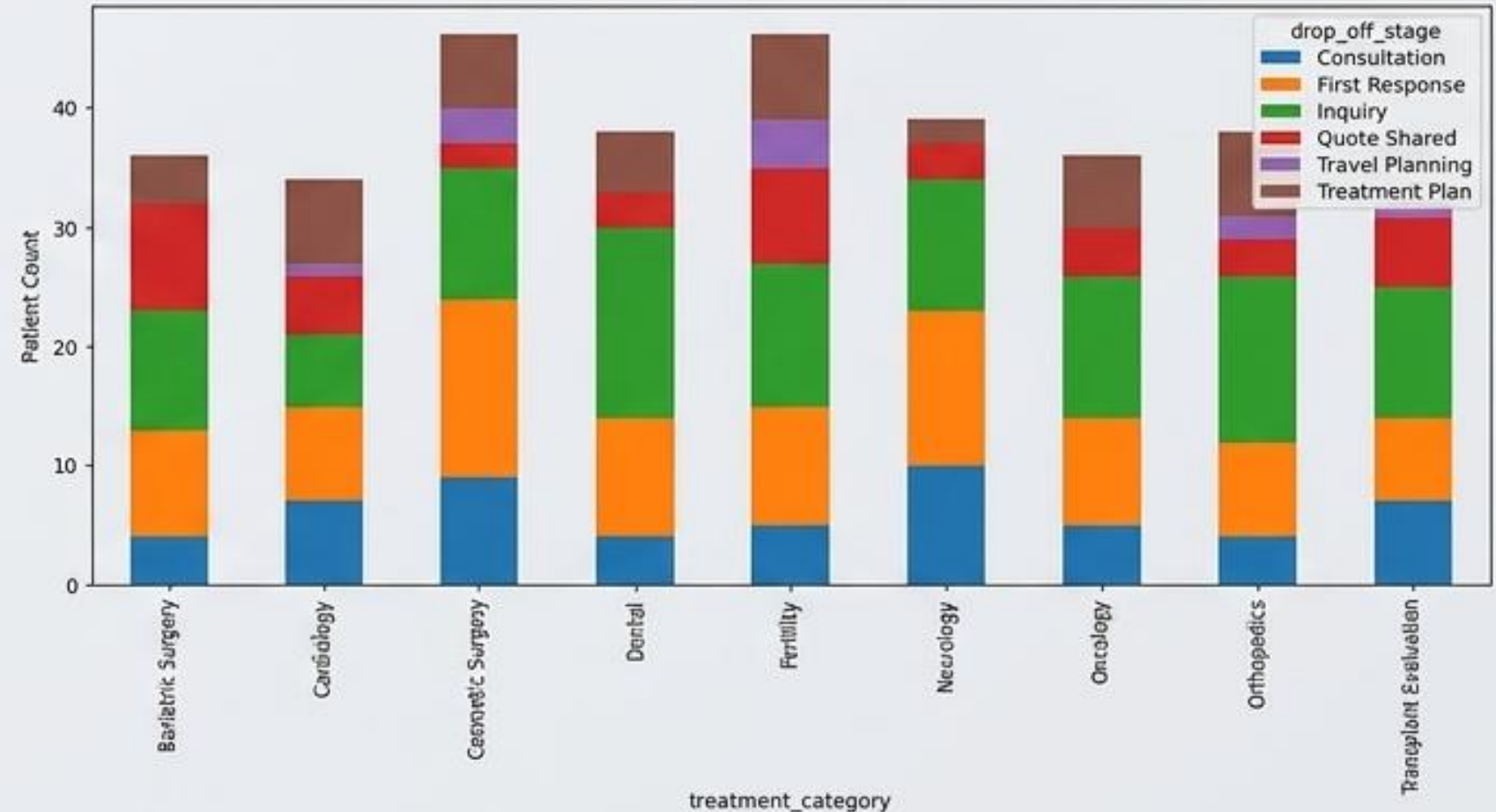
Diagnosing the Leaks by Treatment Category

Overall Friction Points



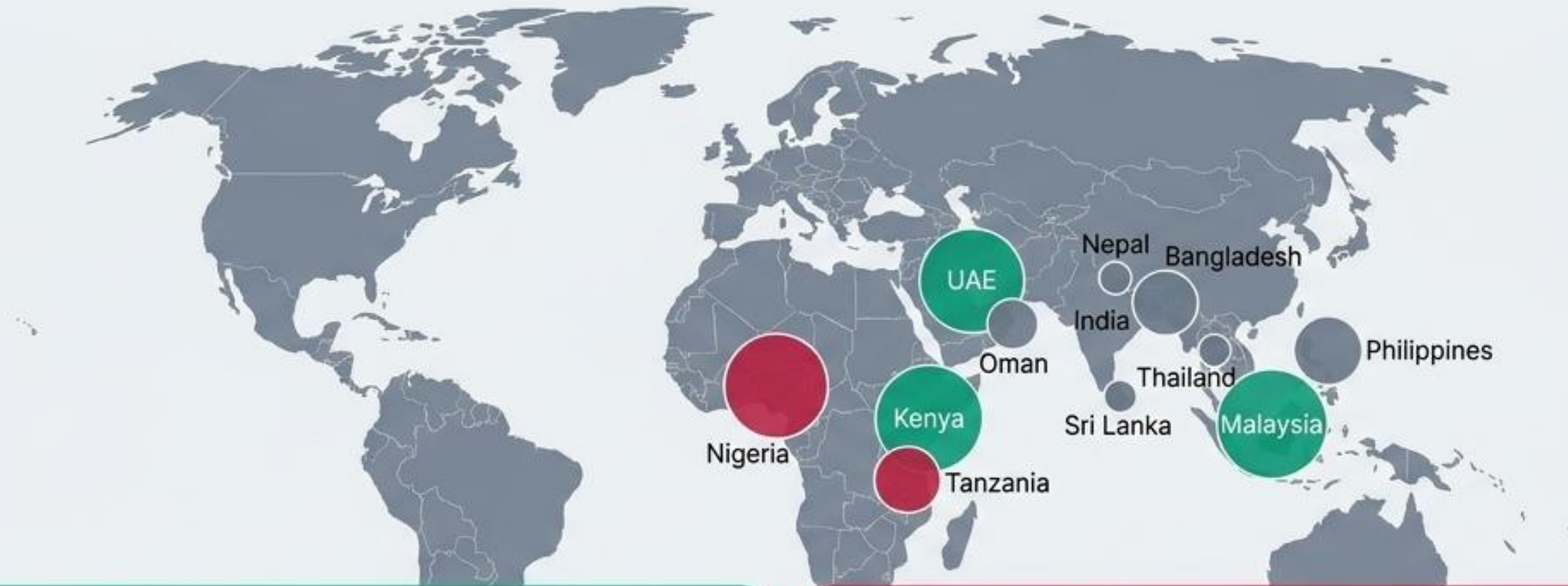
The highest friction points remain at the initial **Inquiry** and **First Response** stages.

Drop-offs by Treatment Category



- **Dental & Orthopaedics:** Suffer severe drop-offs at the initial Inquiry stage (16 and 14 patients lost, respectively), indicating potential price shock or misaligned initial marketing expectations.
- **Cosmetic Surgery:** Experiences peak drop-offs at the First Response stage (15 patients lost), demanding immediate, high-touch SLA enforcement.

Geographical Benchmarking: The Market Disparity

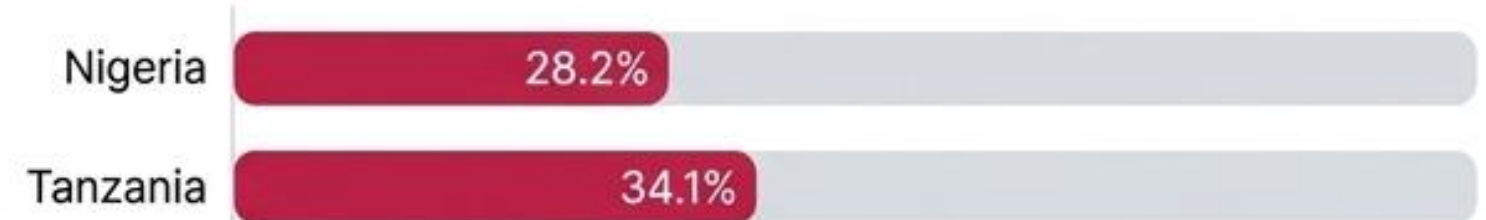


The Leaders (Scale These)



These regions lead the pack in conversion and revenue contribution.

The Laggards (Fix These)



These regions lag significantly behind the baseline.

A one-size-fits-all operational strategy is failing West and East African leads differently. Tailoring market-specific follow-up protocols could instantly recover volume in underperforming regions.

The Cost of Silence: Response Delays Destroy Conversion

Data Annotations:

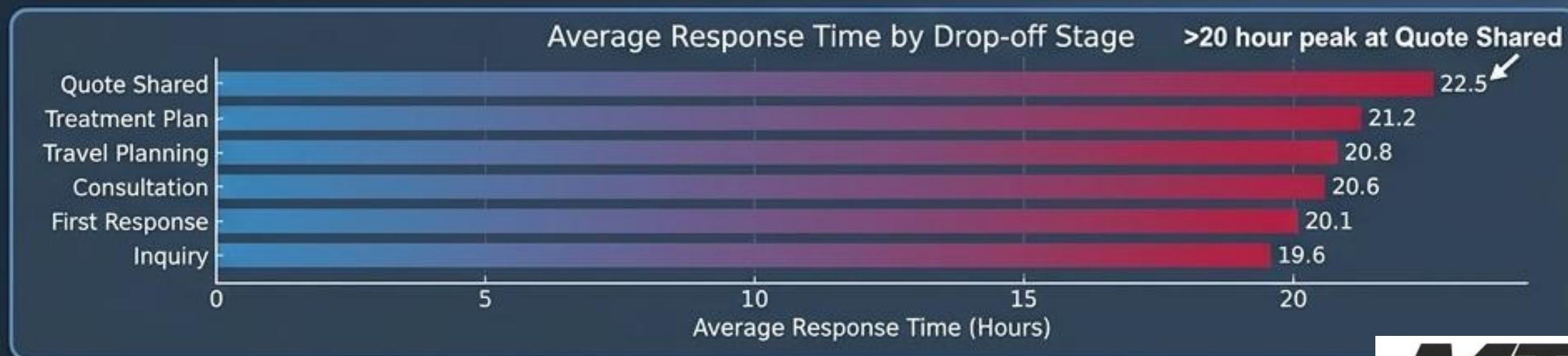
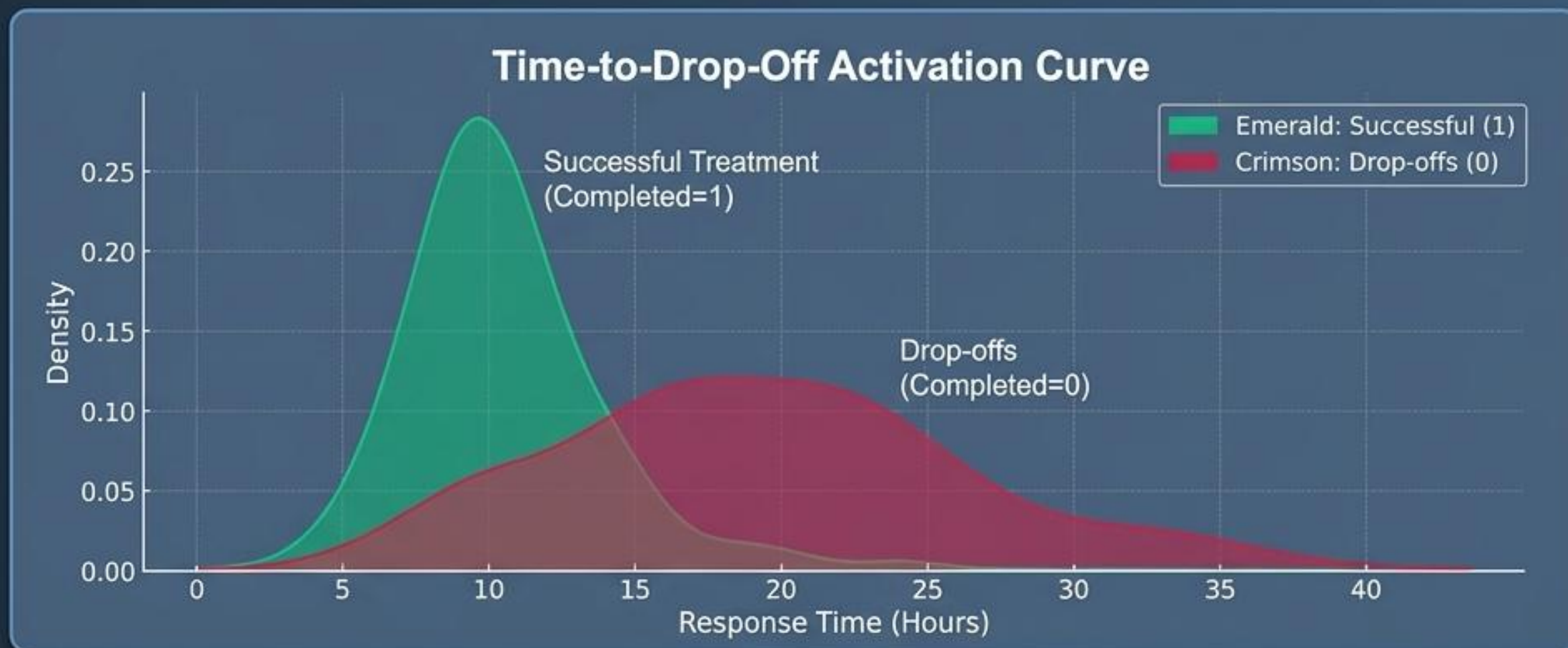
Patients who successfully complete treatment experience heavily clustered, lower response times.

The Breaking Point:

Patients dropping off at the 'Quote Shared' stage experience an average of >20 hours of delay.

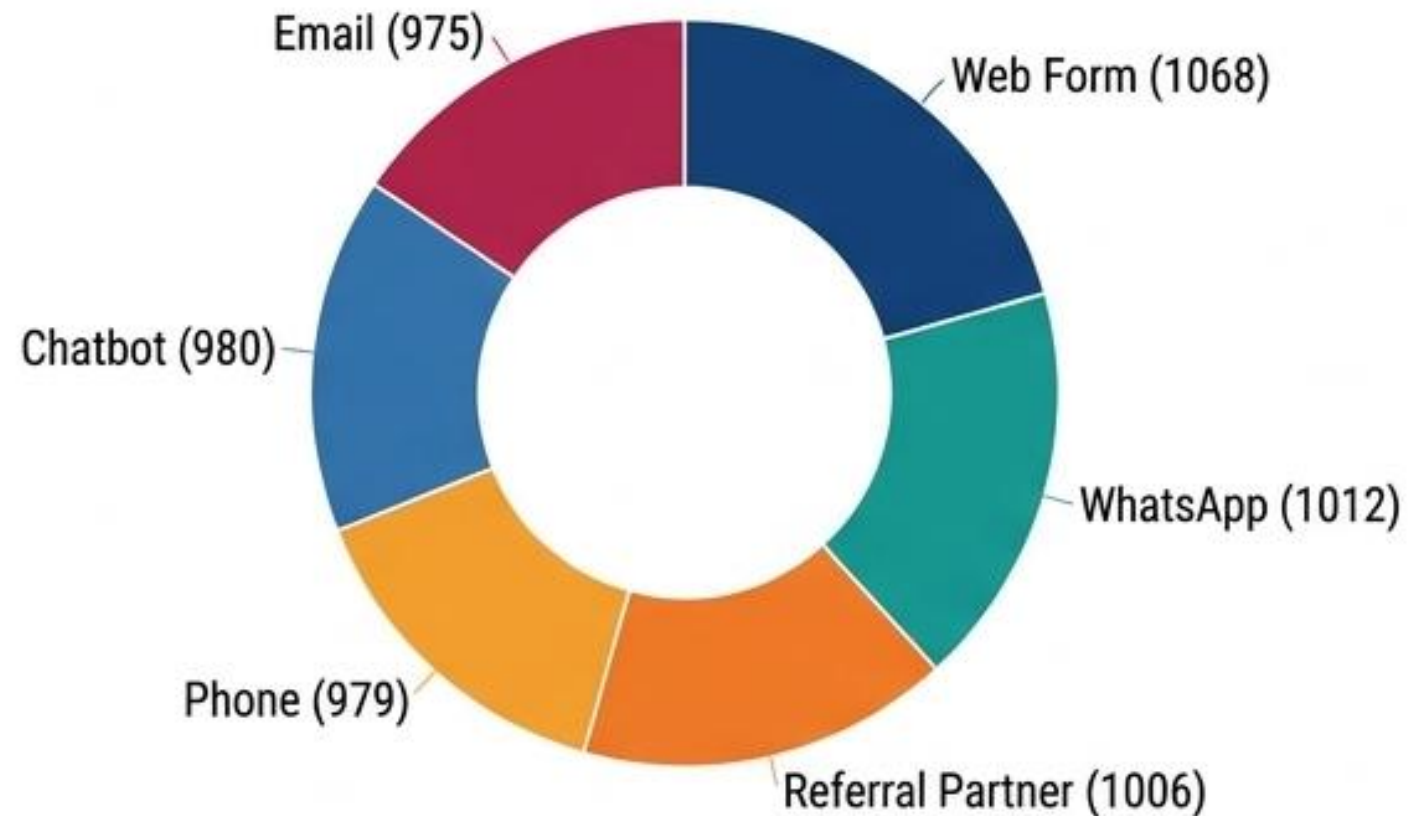
Strategic Callout:

The data proves a strict operational rule: probability of conversion plummets exponentially as response delay increases. Fast SLAs directly inflate engagement scores.

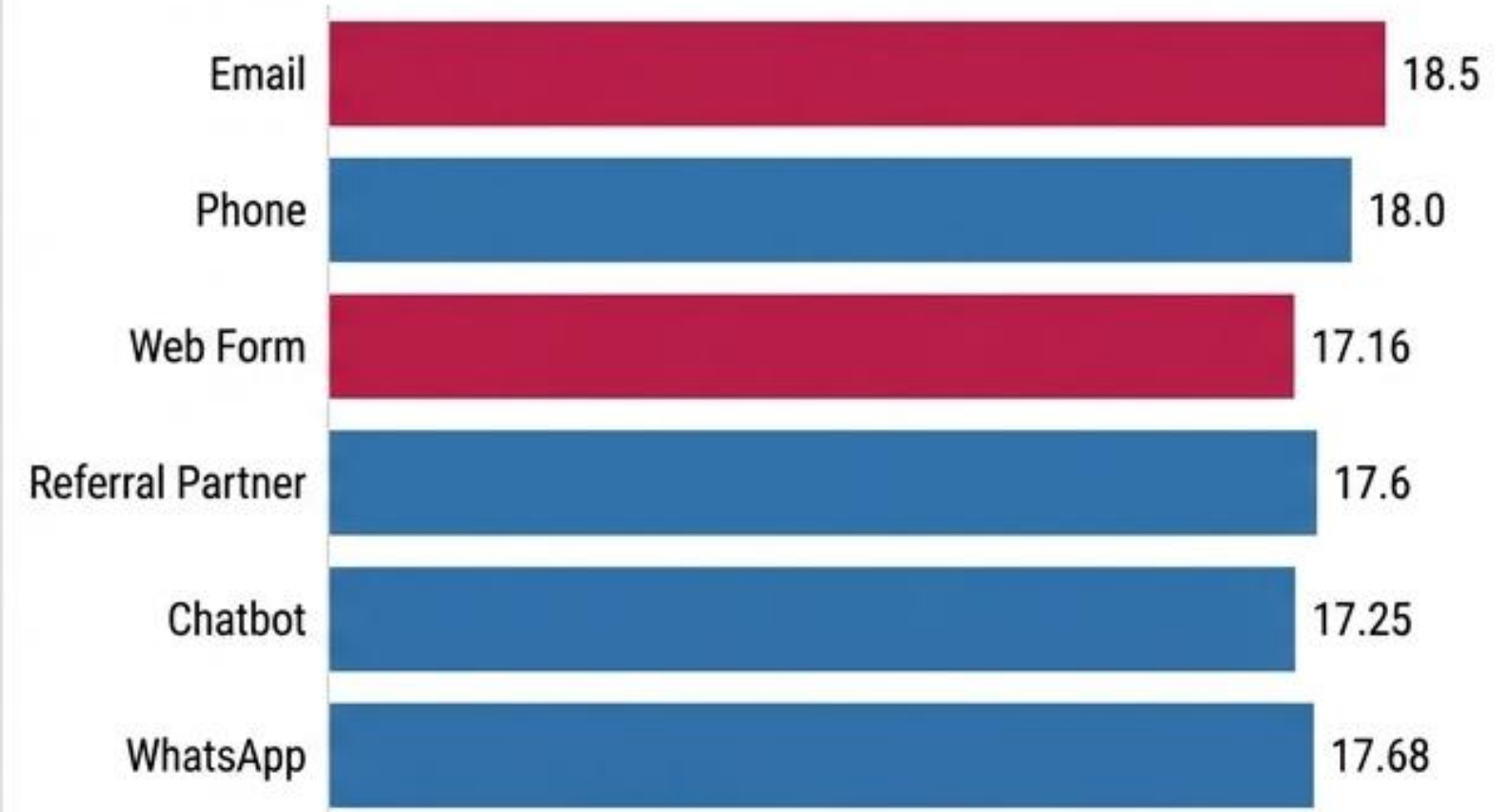


Channel Friction: High Volume vs. Slow Velocity

Channel Volume



Average Delay by Channel (Hours)



The Bottleneck: Web Forms (1,068 interactions) and WhatsApp (1,012 interactions) dominate our lead volume. However, Web Forms average a devastating 17.16 hours of delay.

The Slowest Route: Email remains the absolute slowest channel, averaging 18.5 hours per response.

Strategic Callout: We must aggressively shift volume routing away from slow-batch channels (Email/Web Forms) toward lower-friction, immediate-response channels (Chatbot/WhatsApp) while mandating strict SLA alerts.

Provider Scorecard: Identifying the Network Stars

The Stars (Top Quartile)

PVD-020: 52.7% conversion, ₹13.3M revenue

PVD-008: 50.9% conversion, ₹13.7M revenue

PVD-007: 52.0% conversion, ₹12.2M revenue

These providers combine elite conversion rates with massive revenue generation.

The Laggards (Bottom Quartile)

PVD-028: 23.0% conversion, ₹3.3M revenue

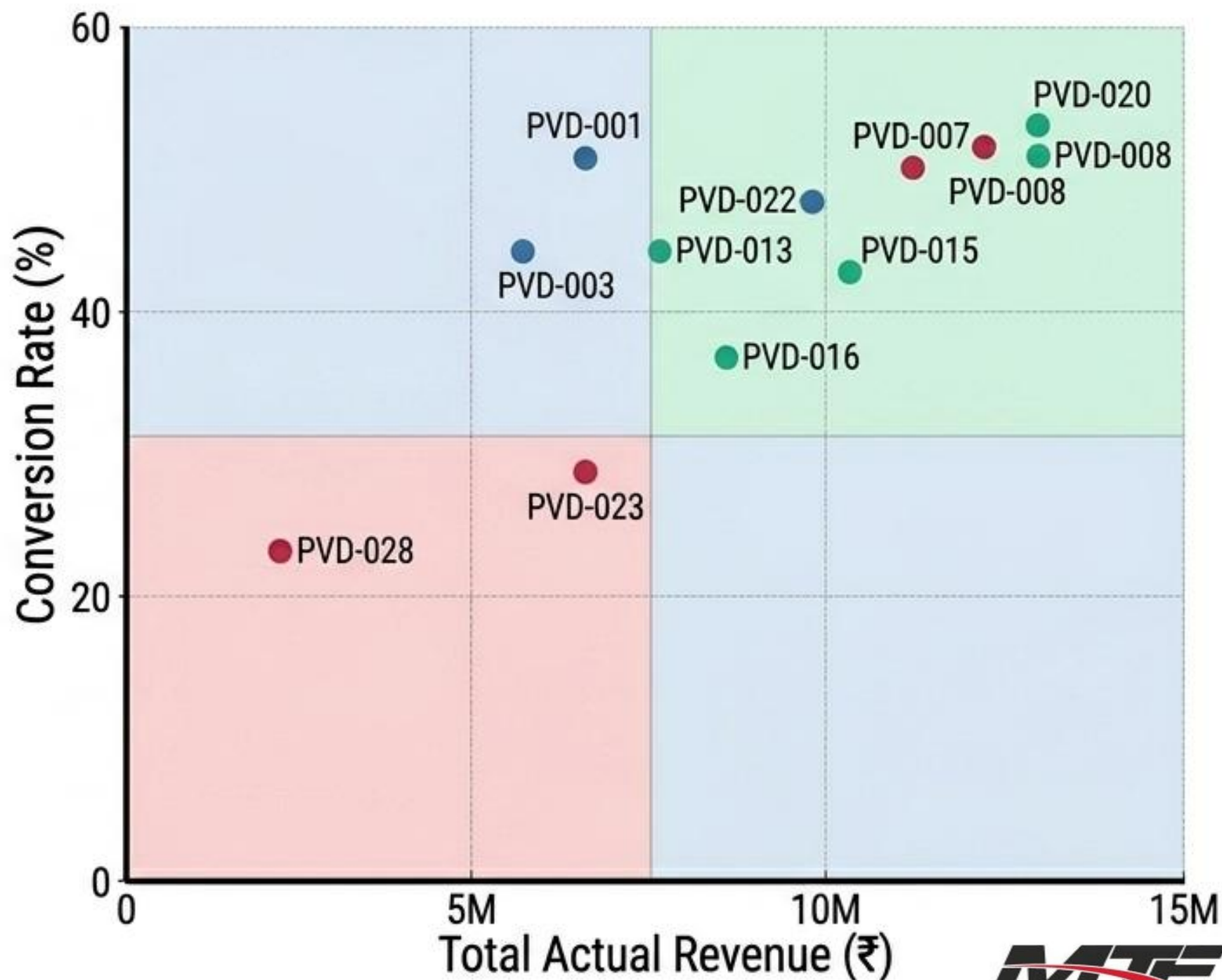
PVD-023: 23.9% conversion

These providers are absorbing leads but failing to convert.

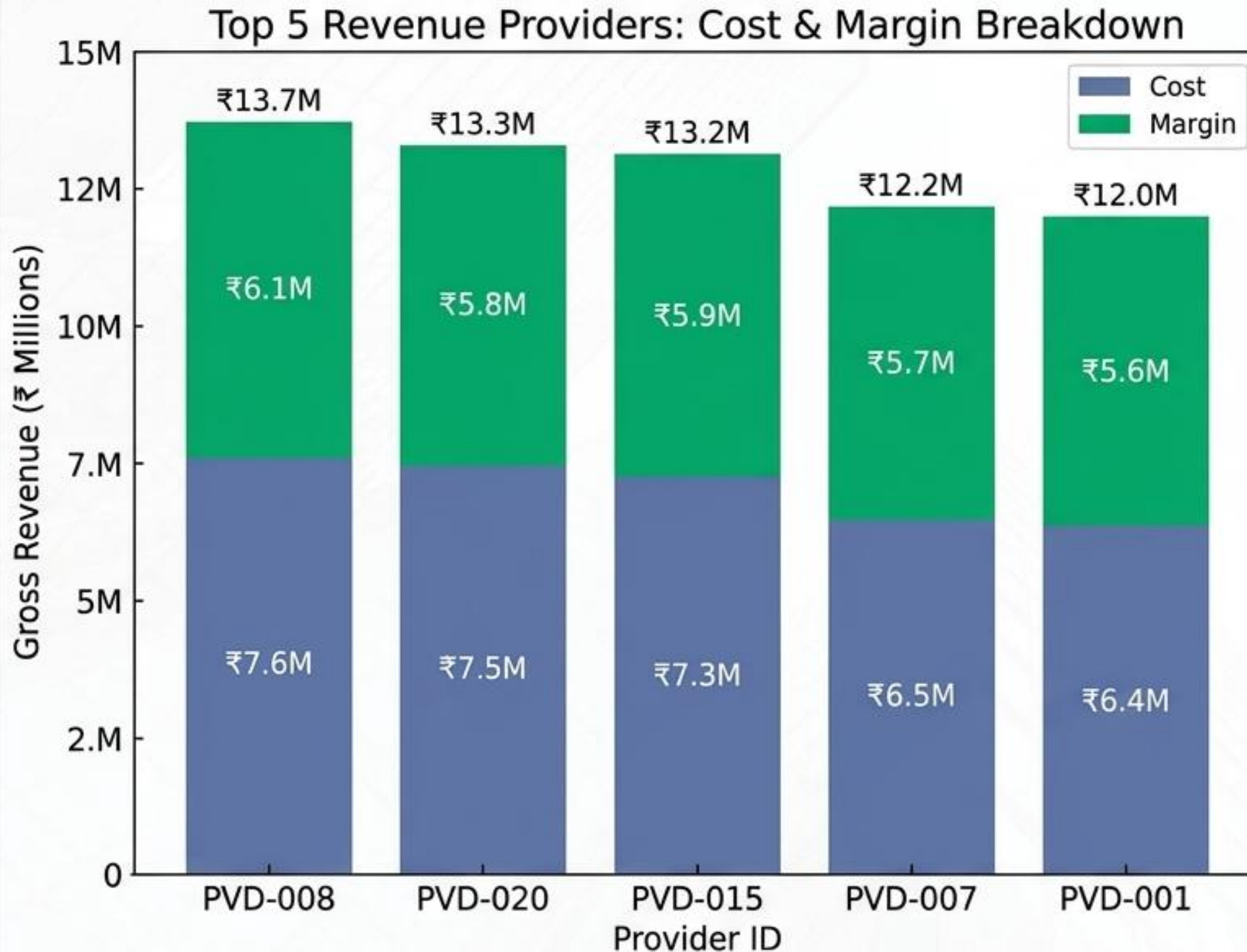
Strategic Callout:

Rerouting just 15% of lead volume from the bottom quartile to the top quartile will geometrically increase total network yield.

The Provider Value Matrix



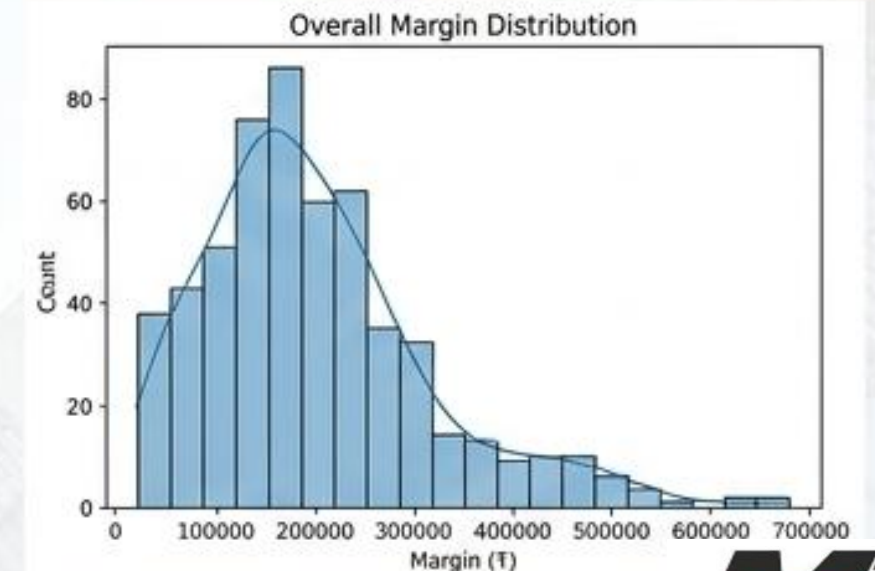
Unit Economics: High Conversion Equals High Margin



Data Highlights:

- PVD-008: Generates ₹13.7M in revenue against ₹7.6M in costs = ₹6.1M Margin.
- PVD-015: ₹13.2M revenue = ₹5.9M Margin.
- PVD-020: ₹13.3M revenue = ₹5.8M Margin.

Strategic Callout: Routing leads to top-tier providers does not just optimise conversion percentages; it drastically impacts the bottom-line profitability of MedTourEasy.



Predictive Intelligence: Anticipating the Drop-Off

Logistic Regression



Accuracy: 1.0

Utility: Foundational baseline proving that operational signatures between won and lost patients are mathematically distinct and non-overlapping.

		Predicted		
		0	1	
Actual	0	182	0	0
	1	0	118	1
		0	1	

Random Forest



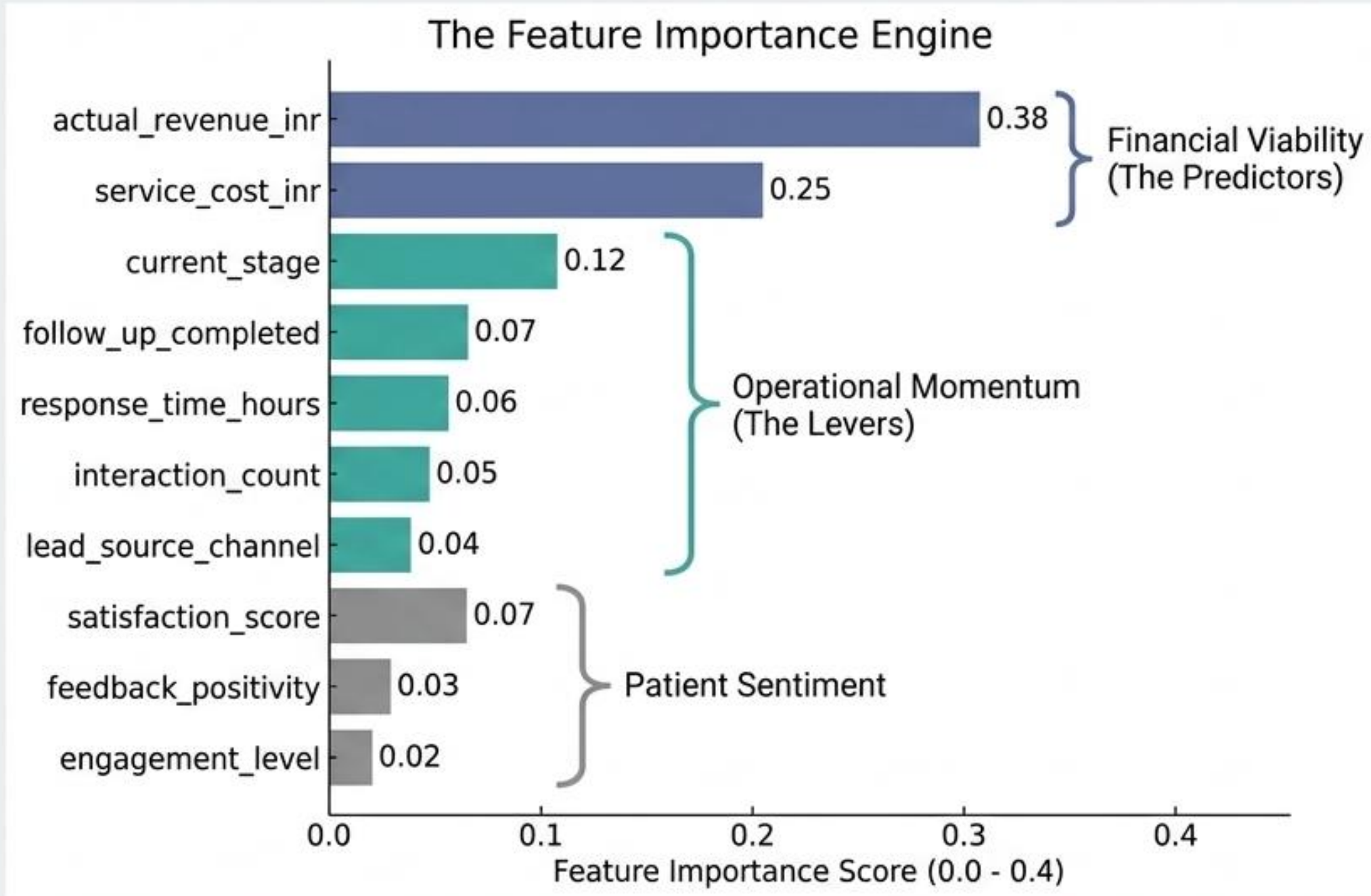
Accuracy: 1.0

Utility: Provides robust, explainable rule-sets. Deployed as an Early-Warning Risk Engine to flag at-risk patients in active pipelines before they disengage.

		Predicted		
		0	1	
Actual	0	182	0	0
	1	0	118	1
		0	1	

The ML Pipeline: We trained supervised learning models on the historical dataset to predict treatment_completed. While perfect historical accuracy is rare, we deploy these models not as infallible crystal balls, but as highly accurate, real-time risk scoring systems.

Decoding Conversion: The Anatomy of a Won Patient



What drives a patient to convert? The ML model ranks importance:



1. Financial Viability (The Predictors)

actual_revenue_inr (0.38) and service_cost_inr (0.25). Case value and cost mapping are the absolute strongest indicators of eventual completion.



2. Operational Momentum (The Levers)

current_stage (0.12) and follow_up_completed (0.07). These are the variables our coordinators can directly manipulate to force conversion.



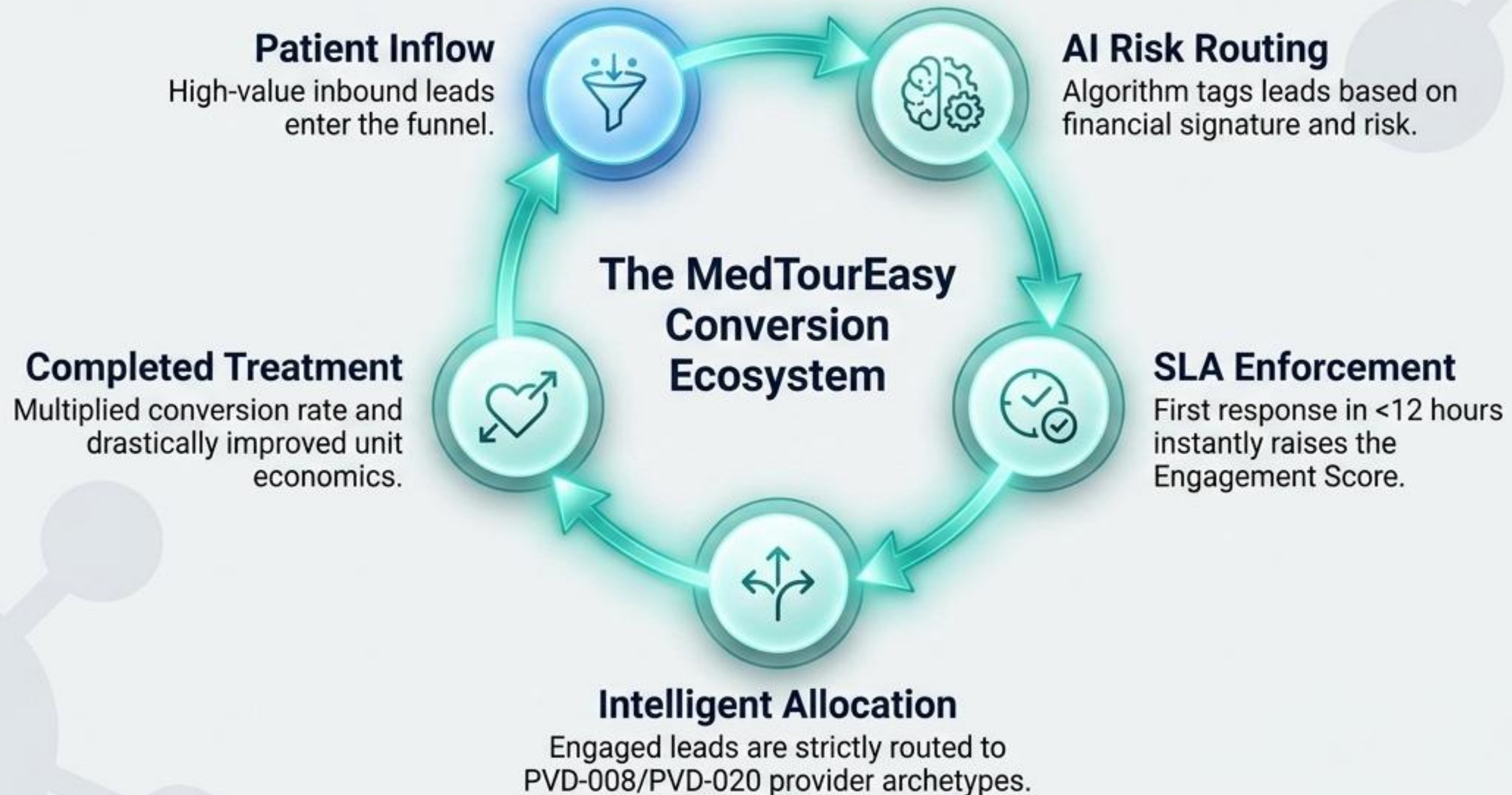
3. Patient Sentiment

satisfaction_score (0.07).

Strategic Callout: We can train our ML engine to dynamically monitor **operational momentum (stage velocity + follow-up density)** to auto-flag accounts slipping into high-risk profiles.

Synthesis: The MedTourEasy Conversion Ecosystem

The Connected Blueprint: These insights do not exist in silos.



Deployed Asset: The Executive SQL Dashboard

Technical Success

The raw fragmented data has been governed, cleaned, and deployed into a 5-table SQLite database.

Real-Time Capability

The infrastructure now supports an automated Python/SQL management dashboard tracking real-time funnel metrics, country benchmarks, and provider KPIs.

Impact

Management is no longer blind to the funnel; operations are now continuously measurable.



Strategic Imperatives & Next Steps

1

Operationalise SLAs (Immediate)

- Implement strict <12hr response mandates for all Web Form and Email inquiries.
- Shift early-funnel volume toward WhatsApp and Chatbots.

2

Provider Prioritisation (Next 30 Days)

- Restructure lead routing.
- High-intent international leads must exclusively route to the top quartile of providers based on the Provider Value Matrix (prioritising PVD-008 and PVD-020).

3

Deploy Predictive Interventions (Next 90 Days)

- Integrate the Random Forest risk-scoring pipeline into the CRM.
- Flag 'at-risk' patients to coordinators on daily dashboards based on delay thresholds and follow-up density.